

RosuZet® (EZETIMIBE 10MG/ ROSUVASTATIN 10MG) FILM COATED TABLET

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Cholesterol Absorption Inhibitor/ HMG-CoA Reductase Inhibitor**1. INDICATIONS AND USAGE*****Primary Hypercholesterolemia***

Ezetimibe + Rosuvastatin (ROSUZET) is indicated as adjunctive therapy to diet for the reduction of elevated total cholesterol (total-C), low-density lipoprotein cholesterol (LDL-C), apolipoprotein B (Apo B), triglycerides (TG), and non-high-density lipoprotein cholesterol (non-HDL-C), and to increase high-density lipoprotein cholesterol (HDL-C) in patients with primary (heterozygous familial and non-familial) hypercholesterolemia or mixed hyperlipidemia.

Homozygous Familial Hypercholesterolemia (HoFH)

Ezetimibe + Rosuvastatin (ROSUZET) is indicated for the reduction of elevated total-C and LDL-C levels in patients with HoFH. Patients may also receive adjunctive treatments (e.g., LDL apheresis).

2. DOSAGE AND ADMINISTRATION**2.1 General**

The patient should be on an appropriate lipid-lowering diet and should continue on this diet during treatment with Ezetimibe + Rosuvastatin (ROSUZET). The dosage should be individualized according to the baseline LDL-C level, the recommended goal of therapy, and the patient's response. Ezetimibe + Rosuvastatin (ROSUZET) can be administered as a single dose at any time of the day, with or without food.

2.2 Adults***Primary Hypercholesterolemia***

The usual start dose is 10/10 mg once daily. Use Ezetimibe 10mg with Rosuvastatin 5mg for special populations if needed. For patients with severe hypercholesterolemia (including heterozygous familial hypercholesterolaemia) or those with aggressive lipid targets, a start dose of 10/20 mg may be considered. Not all strengths are available locally.

Homozygous Familial Hypercholesterolemia

For patients with homozygous familial hypercholesterolemia a start dose of 10/20 mg once a day is recommended. Not all strengths are available locally.

2.3 Use in Pediatric Patients

Treatment with Ezetimibe + Rosuvastatin (ROSUZET) is not recommended.

2.4 Use in the Elderly

No dosage adjustment is required for elderly patients.

2.5 Use in Patients with Renal Insufficiency

No dosage adjustment is required for patients with mild to moderate renal impairment. For patients with severe renal impairment ($CL_{Cr} < 30$ mL/min/1.73 m²) not on hemodialysis, the dose should be started at Ezetimibe 10mg with Rosuvastatin 5mg once daily and not exceed 10/10 mg once daily [See 10 CLINICAL PHARMACOLOGY, 10.3 Pharmacokinetics].

2.6 Use in Patients with Hepatic Insufficiency

No dosage adjustment is required for patients with mild hepatic impairment (Child-Pugh score 5 to 6). Treatment with Ezetimibe + Rosuvastatin (ROSUZET) is not recommended in patients with moderate (Child-Pugh score 7 to 9) or severe (Child-Pugh score >9) liver dysfunction [see 10 CLINICAL PHARMACOLOGY, 10.3 Pharmacokinetics].

2.7 Coadministration with Bile Acid Sequestrants

Dosing of Ezetimibe + Rosuvastatin (ROSUZET) should occur either ≥ 2 hours before or ≥ 4 hours after administration of a bile acid sequestrant.

2.8 Dosage in patients with pre-disposing factors to myopathy

The recommended start dose of rosuvastatin is 5 mg in patients with pre-disposing factors to myopathy.

2.9 Concomitant Therapy

Rosuvastatin is a substrate of various transporter proteins (e.g. OATP1 B1 and BCRP). The risk of myopathy (including, rhabdomyolysis) is increased when Ezetimibe + Rosuvastatin (ROSUZET) is administered concomitantly with certain medicinal products that may increase the plasma concentration of rosuvastatin due to interactions with these transporter proteins (e.g. ciclosporin and certain protease inhibitors including combinations of ritonavir with atazanavir, lopinavir, and/or tipranavir. It is recommended that prescribers consult the relevant product information when considering administration of such products together with Ezetimibe + Rosuvastatin (ROSUZET). Whenever possible, alternative medications should be considered, and if necessary, consider temporarily discontinuing Ezetimibe + Rosuvastatin (ROSUZET) therapy. In situations where co-administration of these "medicinal products with Ezetimibe + Rosuvastatin (ROSUZET) is unavoidable, the benefit and the risk of concurrent treatment and Ezetimibe + Rosuvastatin (ROSUZET) dosing adjustments should be carefully considered [see 4 WARNINGS AND PRECAUTIONS and 5 DRUG INTERACTIONS AND OTHER FORMS OF INTERACTIONS].

2.10 Race

A starting dose of Ezetimibe 10mg with Rosuvastatin 5mg should be considered for Asian patients. Increased plasma concentration of rosuvastatin has been seen in Asian subjects [see 4 WARNINGS AND PRECAUTIONS, 4.10 Race; 10 CLINICAL PHARMACOLOGY, 10.3 Pharmacokinetics]. The increased systemic exposure should be taken into consideration when treating Asian patients whose hypercholesterolaemia is not adequately controlled at doses up to 10/20 mg/day.

2.11 Genetic polymorphisms

Genotypes of SLCO1 B1 (OATP1 B1) c.521CC and ABCG2 (BCRP) c.421 AA have been shown to be associated with an increase in rosuvastatin exposure (AUC)

compared to SLCO1 B1 c.521 TT and ABCG2 c.421 CC. For patients known to have the c.521 CC or c.421 AA genotype, a maximum once daily dose of 10/20 mg of Ezetimibe + Rosuvastatin (ROSUZET) is recommended [see 5 *DRUG INTERACTIONS AND OTHER FORMS OF INTERACTIONS*, 5.2 *Rosuvastatin* and 10 *CLINICAL PHARMACOLOGY*, 10.3 *Pharmacokinetics*].

3. CONTRAINDICATIONS

- Ezetimibe + Rosuvastatin (ROSUZET) is contraindicated in patients with hypersensitivity to ezetimibe, rosuvastatin, or any of its inactive ingredients.
- Ezetimibe + Rosuvastatin (ROSUZET) is contraindicated in patients with active liver disease.
- Ezetimibe + Rosuvastatin (ROSUZET) is contraindicated during pregnancy, while breast-feeding and in women of child-bearing potential not using appropriate contraceptive measures [see 6 *USE IN SPECIFIC POPULATIONS*, 6.1 *Pregnancy* and 6.2 *Nursing Mothers*].
- Ezetimibe + Rosuvastatin (ROSUZET) is contraindicated in patients receiving concomitant cyclosporine.

4. WARNINGS AND PRECAUTIONS

4.1 Skeletal Muscle Effects

As with other HMG-CoA reductase inhibitors, effects on skeletal muscle, e.g. myalgia, myopathy and, rarely, rhabdomyolysis, have been reported in patients treated with rosuvastatin. As with other HMG-CoA reductase inhibitors, the reporting rate for rhabdomyolysis in post-marketing use is higher at the highest marketed dose. Patients who develop any signs or symptoms suggestive of myopathy should have their CK levels measured. Ezetimibe + Rosuvastatin (ROSUZET) therapy should be discontinued if CK levels are markedly elevated ($> 10 \times \text{ULN}$) or if myopathy is diagnosed or suspected. There have been very rare reports of an immune-mediated necrotising myopathy clinically characterized by persistent proximal muscle weakness and elevated serum creatine kinase during treatment or following discontinuation of statins, including rosuvastatin. Additional neuromuscular and serologic testing may be necessary. Treatment with immunosuppressive agents may be required.

In rosuvastatin trials there was no evidence of increased skeletal muscle effects when rosuvastatin was dosed with any concomitant therapy. However, an increase in the incidence of myositis and myopathy has been seen in patients receiving other HMG-CoA reductase inhibitors together with ciclosporin, fibric acid derivatives (including gemfibrozil), nicotinic acid, azole antifungals and macrolide antibiotics.

Ezetimibe + Rosuvastatin (ROSUZET) should be prescribed with caution in patients with pre-disposing factors for myopathy, such as, renal impairment, advanced age and hypothyroidism, or situations where an increase in plasma levels may occur [see 5 *DRUG INTERACTIONS AND OTHER FORMS OF INTERACTIONS*, 5.2 *Rosuvastatin*].

Ezetimibe + Rosuvastatin (ROSUZET) should be temporarily withheld in any patient

with an acute serious condition suggestive of myopathy or predisposing to the development of renal failure secondary to rhabdomyolysis (e.g. sepsis, hypotension, major surgery, trauma, severe metabolic, endocrine and electrolyte disorders; or uncontrolled seizures).

Reports of myopathy and/or rhabdomyolysis have been observed with HMG-CoA reductase inhibitors coadministered with daptomycin. Caution should be used when prescribing HMG-CoA reductase inhibitors with daptomycin, as either agent can cause myopathy and/or rhabdomyolysis when given alone. Consideration should be given to suspending Ezetimibe + Rosuvastatin (ROSUZET) temporarily in patients taking daptomycin [see 5 DRUG INTERACTIONS AND OTHER FORMS OF INTERACTIONS, 5.2 Rosuvastatin].

4.2 Myasthenia Gravis/Ocular Myasthenia

In few cases, statins have been reported to induce de novo or aggravate pre-existing myasthenia gravis or ocular myasthenia [see 7 ADVERSE REACTIONS]. ROSUZET should be discontinued in case of aggravation of symptoms. Recurrences when the same or a different statin was (re-) administered have been reported.

4.3 Liver Enzyme Abnormalities

In controlled coadministration trials in patients receiving ezetimibe with a statin, consecutive transaminase elevations (≥ 3 X the upper limit of normal [ULN]) have been observed. [See 7 ADVERSE REACTIONS]

In the IMProved Reduction of Outcomes: Vytorin Efficacy International Trial (IMPROVE-IT), 18,144 patients with CHD were randomized to receive ezetimibe/simvastatin 10/40 mg daily (n=9067) or simvastatin 40 mg daily (n= 9077). During a median follow-up of 6.0 years, the incidence of consecutive elevations of transaminases (≥ 3 X ULN) was 2.5% for ezetimibe/simvastatin and 2.3% for simvastatin.

In a controlled clinical study in which over 9000 patients with chronic kidney disease were randomized to receive Ezetimibe (EZETROL™) Tablet 10 mg combined with simvastatin 20 mg daily (n=4650) or placebo (n=4620) (median follow-up period of 4.9 years), the incidence of consecutive elevations of transaminases (≥ 3 X ULN) was 0.7% for Ezetimibe (EZETROL™) Tablet combined with simvastatin and 0.6% for placebo.

Ezetimibe + Rosuvastatin (ROSUZET) should be used with caution in patients who consume excessive quantities of alcohol and/or have a history of liver disease.

4.4 Hepatic Impairment

Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe hepatic impairment, Ezetimibe + Rosuvastatin (ROSUZET) is not recommended in these patients.

4.5 Gemfibrozil

Due to an observed increased risk of myopathy/rhabdomyolysis, combination therapy with Ezetimibe + Rosuvastatin (ROSUZET) and gemfibrozil should be avoided [see 5 DRUG INTERACTIONS AND OTHER FORMS OF INTERACTIONS]. If used together, the dose of Ezetimibe + Rosuvastatin (ROSUZET) should not exceed 10/10 mg once daily.

4.6 Fenofibrate

If cholelithiasis is suspected in a patient receiving Ezetimibe + Rosuvastatin (ROSUZET) and fenofibrate, gallbladder studies are indicated and alternative lipid-lowering therapy should be considered [see the product labeling for fenofibrate and fenofibric acid].

4.7 Other Fibrates

The coadministration of ezetimibe with fibrates other than fenofibrate has not been studied. Therefore, coadministration of Ezetimibe + Rosuvastatin (ROSUZET) and fibrates (other than fenofibrate) is not recommended [see 5 *DRUG INTERACTIONS AND OTHER FORMS OF INTERACTIONS*].

4.8 Anticoagulants

If Ezetimibe + Rosuvastatin (ROSUZET) is added to warfarin, another coumarin anticoagulant, or fluindione, the International Normalized Ratio (INR) should be appropriately monitored [see 5 *DRUG INTERACTIONS AND OTHER FORMS OF INTERACTIONS*].

4.9 Diabetes Mellitus

As with other HMG-CoA reductase inhibitors, increases in HbA1c and serum glucose levels have been observed in patients treated with rosuvastatin, and in some instances these increases may exceed the threshold for the diagnosis of diabetes mellitus, primarily in patients already at high risk for developing diabetes [see 7 *ADVERSE REACTIONS*; 10 *CLINICAL PHARMACOLOGY*].

4.10 Race

Pharmacokinetic studies show an increase in rosuvastatin exposure in Asian subjects compared with Caucasians [see 2 *DOSAGE AND ADMINISTRATION*, 2.9 *Race*; 10 *CLINICAL PHARMACOLOGY*, 10.3 *Pharmacokinetics*].

5. DRUG INTERACTIONS AND OTHER FORMS OF INTERACTIONS

No clinically significant pharmacokinetic interaction was seen when ezetimibe was coadministered with rosuvastatin.

Multiple mechanisms may contribute to potential interactions with HMG Co-A reductase inhibitors. Drugs or herbal products that inhibit certain enzymes and/or transporter (e.g. OATP1B) pathways may increase rosuvastatin plasma concentrations and may lead to an increased risk of myopathy/rhabdomyolysis.

Consult the prescribing information of all concomitantly used drugs to obtain further information about their potential interactions with rosuvastatin and/or the potential for enzyme or transporter alterations and possible adjustments to dose and regimens.

5.1 Ezetimibe

In preclinical studies, it has been shown that ezetimibe does not induce cytochrome P450 drug metabolizing enzymes. No clinically significant pharmacokinetic interactions have been observed between ezetimibe and drugs known to be metabolized by

cytochromes P450 1A2, 2D6, 2C8, 2C9, and 3A4, or N-acetyltransferase.

Ezetimibe had no effect on the pharmacokinetics of dapsone, dextromethorphan, digoxin, oral contraceptives (ethinyl estradiol and levonorgestrel), glipizide, tolbutamide, or midazolam during co-administration. Cimetidine, co-administered with ezetimibe, had no effect on the bioavailability of ezetimibe.

Antacids: Concomitant antacid administration decreased the rate of absorption of ezetimibe but had no effect on the bioavailability of ezetimibe. This decreased rate of absorption is not considered clinically significant.

Cholestyramine: Concomitant cholestyramine administration decreased the mean AUC of total ezetimibe (ezetimibe + ezetimibe glucuronide) approximately 55 %. The incremental LDL- C reduction due to adding ezetimibe to cholestyramine may be lessened by this interaction.

Cyclosporine: In a study of eight post-renal transplant patients with creatinine clearance of >50 mL/min on a stable dose of cyclosporine, a single 10-mg dose of ezetimibe resulted in a 3.4-fold (range 2.3- to 7.9-fold) increase in the mean AUC for total ezetimibe compared to a healthy control population from another study (n=17). In a different study, a renal transplant patient with severe renal insufficiency (creatinine clearance of 13.2 mL/min/1.73 m²) who was receiving multiple medications, including cyclosporine, demonstrated a 12- fold greater exposure to total ezetimibe compared to concurrent controls. In a two-period crossover study in twelve healthy subjects, daily administration of 20 mg ezetimibe for 8 days with a single 100- mg dose of cyclosporine on Day 7 resulted in a mean 15% increase in cyclosporine AUC (range 10% decrease to 51% increase) compared to a single 100- mg dose of cyclosporine alone.

Fibrates: The safety and effectiveness of ezetimibe co-administered with fenofibrate have been evaluated in a clinical study; co-administration of ezetimibe with other fibrates has not been studied. Fibrates may increase cholesterol excretion into the bile, leading to cholelithiasis. In a preclinical study in dogs, ezetimibe increased cholesterol in the gallbladder bile. Although the relevance of this preclinical finding to humans is unknown, co-administration of ezetimibe with fibrates (other than fenofibrate) is not recommended until use in patients is studied. Concurrent use of fibrates may cause severe myositis and myoglobinuria.

Fenofibrate: In a pharmacokinetic study, concomitant fenofibrate administration increased total ezetimibe concentrations approximately 1.5- fold. This increase is not considered clinically significant.

Gemfibrozil: In a pharmacokinetic study, concomitant gemfibrozil administration increased total ezetimibe concentrations approximately 1.7- fold. This increase is not considered clinically significant. No clinical data are available.

Statins: No clinically significant pharmacokinetic interactions were seen when ezetimibe was co- administered with atorvastatin, simvastatin, pravastatin, lovastatin, fluvastatin, or rosuvastatin.

Anticoagulants: Concomitant administration of ezetimibe (10 mg once daily) had no significant effect on bioavailability of warfarin and prothrombin time in a study of twelve healthy adult males. There have been post-marketing reports of increased International Normalized Ratio in patients who had ezetimibe added to warfarin or fluindione. Most of these patients were also on other medications [see 4 WARNINGS AND PRECAUTIONS, 4.8 Anticoagulants].

5.2 Rosuvastatin

In vitro and *in vivo* data indicate that rosuvastatin has no clinically significant cytochrome P450 interactions (as a substrate, inhibitor or inducer). Rosuvastatin is a substrate for certain transporter proteins including the hepatic uptake transporter OATP1B1 and efflux transporter BCRP. Concomitant administration of rosuvastatin with medicinal products that are inhibitors of these transporter proteins may result increased rosuvastatin plasma concentrations and an increased risk of myopathy [see Table 1; 2 DOSAGE AND ADMINISTRATION; 4 WARNINGS AND PRECAUTIONS].

Table 1 Effect of co-administered medicinal products on rosuvastatin exposure (AUC; in order of decreasing magnitude) from published clinical trials		
Interacting drug dose regimen	Rosuvastatin dose regimen	Change in rosuvastatin AUC
Cyclosporin 75 mg BID to 200 mg BID, 6 months	10 mg OD, 10 days	7.1-fold ↑
Atazanavir 300 mg/ritonavir 100 mg OD, 8 days	10 mg, single dose	3.1-fold ↑
Simeprevir 150 mg OD, 7 days	10 mg, single dose	2.8-fold ↑
Lopinavir 400 mg/ritonavir 100 mg BID, 17 days	20 mg OD, 7 days	2.1-fold ↑
Clopidogrel 300 mg loading, followed by 75 mg at 24 hours	20 mg, single dose	2-fold ↑
Gemfibrozil 600 mg BID, 7 days	80 mg, single dose	1.9-fold ↑
Eltrombopag 75 mg OD, 5 days	10 mg, single dose	1.6-fold ↑
Darunavir 600 mg/ritonavir 100 mg BID, 7 days	10 mg OD, 7 days	1.5-fold ↑
Tipranavir 500 mg/ritonavir 200 mg BID, 11 days	10 mg, single dose	1.4-fold ↑
Dronedarone 400 mg BID	Not available	1.4-fold ↑
Itraconazole 200 mg OD, 5 days	10 mg or 80 mg, single dose	1.4-fold ↑
Ezetimibe 10 mg OD, 14 days	10 mg, OD, 14 days	1.2-fold ↑
Fosamprenavir 700 mg/ritonavir	10 mg, single dose	

100 mg BID, 8 days		↔
Aleglitazar 0.3 mg, 7 days	40 mg, 7 days	↔
Silymarin 140 mg TID, 5 days	10 mg, single dose	↔
Fenofibrate 67 mg TID, 7 days	10 mg, 7 days	↔
Rifampin 450 mg OD, 7 days	20 mg, single dose	↔
Ketoconazole 200 mg BID, 7 days	80 mg, single dose	↔
Fluconazole 200 mg OD, 11 days	80 mg, single dose	↔
Erythromycin 500 mg QID, 7 days	80 mg, single dose	20% ↓
Baicalin 50 mg TID, 14 days	20 mg, single dose	47% ↓

Interaction requiring rosuvastatin dose adjustments (see also Table 1):

When it is necessary to co-administer Ezetimibe + Rosuvastatin (ROSUZET) with other medicinal products known to increase exposure to rosuvastatin, doses of Ezetimibe + Rosuvastatin (ROSUZET) should be adjusted. It is recommended that prescribers consult the relevant product information when considering administration of such products together with Ezetimibe + Rosuvastatin (ROSUZET). Start with a once daily dose of Ezetimibe 10mg with Rosuvastatin 5mg if the expected increase in exposure (AUC) is approximately 2-fold or higher. The maximum daily dose of Ezetimibe + Rosuvastatin (ROSUZET) should be adjusted so that the expected rosuvastatin exposure would not likely exceed that of a 10/40 mg daily dose of Ezetimibe + Rosuvastatin (ROSUZET) taken without interacting medicinal products, for example a 10/5 mg dose of Ezetimibe + Rosuvastatin with ciclosporin (7.1-fold increase in exposure), a 10/10 mg dose of Ezetimibe + Rosuvastatin (ROSUZET) with ritonavir/atazanavir combination (3.1-fold increase) and a 10/20 mg dose of Ezetimibe + Rosuvastatin (ROSUZET) with gemfibrozil (1.9-fold increase). Not all strengths are available locally.

Other interacting medicinal products

Antacid: The simultaneous dosing of rosuvastatin with an antacid suspension containing aluminum and magnesium hydroxide resulted in a decrease in rosuvastatin plasma concentration of approximately 50%. This effect was mitigated when the antacid was dosed 2 hours after rosuvastatin. The clinical relevance of this interaction has not been studied.

Fusidic Acid: Interaction studies with rosuvastatin and fusidic acid have not been conducted. As with other statins, muscle related events, including rhabdomyolysis, have been reported in post-marketing experience with rosuvastatin and fusidic acid given concurrently. Patients should be closely monitored and temporary suspension of rosuvastatin treatment may be appropriate.

Effect of rosuvastatin on co-administered medicinal products

Warfarin: The pharmacokinetics of warfarin are not significantly affected following co-administration with rosuvastatin. However, as with other HMG-CoA reductase inhibitors, co-administration of rosuvastatin and warfarin may result in a rise in INR compared to warfarin alone. In patients taking vitamin K antagonists monitoring of INR is recommended both at initiation or cessation of therapy with rosuvastatin or following dose adjustment.

Fenofibrates/fibric acid derivatives: Although no pharmacokinetic interaction between rosuvastatin and fenofibrate was observed; a pharmacodynamic interaction may occur. Gemfibrozil, fenofibrate and other fibric acids, including nicotinic acid, may increase the risk of myopathy when given concomitantly with HMG-CoA reductase inhibitors [see 4 WARNINGS AND PRECAUTIONS, 4.5 Gemfibrozil and 4.6 Fenofibrate].

Ciclosporin: Co-administration of rosuvastatin with ciclosporin resulted in no significant changes in ciclosporin plasma concentration.

Other medications:

There was no clinically significant interactions with an oral contraceptive, digoxin, or ezetimibe or fenofibrate.

In clinical studies rosuvastatin was co-administered with antihypertensive agents, antidiabetic agents and hormone replacement therapy. These studies did not produce any evidence of clinically significant adverse interactions.

Daptomycin: The risk of myopathy and/or rhabdomyolysis may be increased by concomitant administration of HMG-CoA reductase inhibitors and daptomycin [see 4 WARNINGS AND PRECAUTIONS].

6. USE IN SPECIFIC POPULATIONS

6.1 Pregnancy

Ezetimibe + Rosuvastatin (ROSUZET)

Ezetimibe + Rosuvastatin (ROSUZET) is contraindicated during pregnancy. Women of child-bearing potential should use appropriate contraceptive measures [see 3 CONTRAINDICATIONS].

Ezetimibe

No clinical data on exposed pregnancies are available. Animal studies of ezetimibe administered alone do not indicate direct or indirect harmful effects with respect to pregnancy, embryonal/fetal development, parturition or postnatal development.

When ezetimibe was given with lovastatin, simvastatin, pravastatin or atorvastatin, no teratogenic effects were observed in embryo-fetal development studies in pregnant rats. In pregnant rabbits, a low incidence of skeletal malformations was observed.

Rosuvastatin

The safety of rosuvastatin during pregnancy and whilst breast-feeding has not been established.

6.2 Nursing Mothers

Ezetimibe + Rosuvastatin (ROSUZET)

Ezetimibe + Rosuvastatin (ROSUZET) is contraindicated in nursing mothers. Because of the potential for serious adverse reactions in a breastfed infant, women who are nursing should not take Ezetimibe + Rosuvastatin (ROSUZET).

Ezetimibe

Studies in rats have shown that ezetimibe is excreted in milk. It is not known whether ezetimibe is excreted into human breast milk.

Rosuvastatin

The safety of rosuvastatin during pregnancy and whilst breast-feeding has not been established.

6.3 Pediatric Use

There are insufficient data for the safe and effective use of Ezetimibe + Rosuvastatin (ROSUZET) in pediatric patients.

7. ADVERSE REACTIONS

7.1 Clinical Trials

Experience Adults

Ezetimibe + Rosuvastatin (ROSUZET)

Ezetimibe + Rosuvastatin (ROSUZET) (as coadministration of ezetimibe and rosuvastatin equivalent to Ezetimibe + Rosuvastatin (ROSUZET)) has been evaluated for safety in a clinical trial with more than 220 patients. Ezetimibe + Rosuvastatin (ROSUZET) was generally well tolerated.

The following uncommon ($\geq 1/1000$, $<1/100$) drug-related adverse experiences were reported in patients taking Ezetimibe + Rosuvastatin (ROSUZET):

Gastrointestinal disorders:

Uncommon: abdominal distension; abdominal pain; constipation; dry mouth; nausea

Skin and subcutaneous tissue disorders:

Uncommon: allergic dermatitis; eczema

Musculoskeletal and connective tissue disorders:

Uncommon: arthralgia; myalgia

Ezetimibe/Rosuvastatin Fixed Dose Combination

The following common ($\geq 1/100$, $<1/10$) or uncommon ($\geq 1/1000$, $<1/100$) drug-related adverse experiences were reported in patients taking a fixed dose combination of ezetimibe/rosuvastatin in a clinical trial with more than 200 Korean patients.

Gastrointestinal disorders:

Uncommon: nausea

Skin and subcutaneous tissue disorders:

Common: pruritus

Uncommon: rash

Investigations:

Uncommon: ALT increased

7.2 Post-marketing Experience and Other Clinical Trial Experience

The following additional adverse reactions have been reported in clinical studies or post-marketing use with ezetimibe (with or without a statin) or rosuvastatin:

Blood and lymphatic system disorders: thrombocytopenia

Immune system disorders: hypersensitivity reactions, including anaphylaxis, angioedema, rash, and urticaria

Endocrine disorders: diabetes mellitus [frequency will depend on the presence or absence of risk factors (fasting blood glucose ≥ 5.6 mmol/L, BMI >30 kg/m², raised triglycerides, history of hypertension)]

Metabolism and nutrition disorders: decreased appetite

Psychiatric disorders: depression, sleep disorders (including insomnia and nightmares)

Nervous system disorders: dizziness; headache; paresthesia; peripheral neuropathy; Frequency 'not known': myasthenia gravis [see 4. WARNINGS AND PRECAUTIONS, 4.2 Aggravation of Myasthenia Gravis/Ocular Myasthenia].

There have been rare postmarketing reports of memory loss associated with statin use. These cognitive issues have been reported for all statins. The reports are generally nonserious, and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median of 3 weeks).

Eye disorders:

Frequency 'not known': ocular myasthenia [see 4. WARNINGS AND PRECAUTIONS, 4.2 Myasthenia Gravis/Ocular Myasthenia].

Vascular disorders: hot flush; hypertension

Respiratory, thoracic, and mediastinal disorders: cough

Gastrointestinal disorders: diarrhea; dyspepsia; flatulence; gastritis; gastroesophageal reflux disease; pancreatitis.

Hepatobiliary disorders: jaundice; hepatitis; increased hepatic transaminase; cholelithiasis; cholecystitis; drug-induced liver injury

Skin and subcutaneous tissue disorders: urticaria; severe cutaneous adverse reactions (SCARs), including Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN), drug reaction with eosinophilic and systemic symptoms (DRESS) and erythema multiforme

Musculoskeletal and connective tissue disorders: back pain; muscle spasms; muscular weakness; myopathy (including myositis) and rhabdomyolysis and [see 4 WARNINGS AND PRECAUTIONS, 4.1 *Skeletal Muscle Effects*]; neck pain; pain in extremity; Immune-mediated necrotizing myopathy

Reproductive system and breast disorders: gynecomastia

General disorders and administration site conditions: asthenia; chest pain; fatigue; pain; peripheral edema

Investigations: AST increased; blood CPK increased; gamma-glutamyltransferase increased; liver function test abnormal

Skeletal Muscle Effects: Rare cases of rhabdomyolysis, which were occasionally associated with impairment of renal function, have been reported with rosuvastatin and with other marketed statins.

Laboratory Effects: As with other HMG-CoA reductase inhibitors, a dose-related increase in liver transaminases and CK has been observed in a small number of patients taking rosuvastatin [See 4 WARNINGS AND PRECAUTIONS, 4.3 *Liver Enzyme Abnormalities and 10 CLINICAL PHARMACOLOGY*]. Increases in HbA1c have also been observed in a small number of patients treated with rosuvastatin. Abnormal urinalysis testing (dipstick-positive proteinuria) has been seen in a small number of patients taking rosuvastatin and other HMG-CoA reductase inhibitors. The protein detected was mostly tubular in origin. In most cases, proteinuria decreases or disappears spontaneously on continued therapy and is not predictive, of acute or progressive renal disease.

Other effects:

In a long-term controlled clinical trial Rosuvastatin was shown to have no harmful effects on the ocular lens.

In rosuvastatin treated patients, there was no impairment of adrenocortical function.

8. OVERDOSAGE

Ezetimibe + Rosuvastatin (ROSUZET)

No specific treatment of overdosage with Ezetimibe + Rosuvastatin (ROSUZET) can be recommended. In the event of an overdose, symptomatic and supportive measures should be employed.

Ezetimibe

In clinical studies, administration of ezetimibe, 50 mg/day to 15 healthy subjects for up to 14 days, 40 mg/day to 18 patients with primary hyperlipidemia for up to 56 days, and 40 mg/day to 27 patients with homozygous sitosterolemia for 26 weeks, was generally well tolerated.

A few cases of overdosage have been reported; most have not been associated with adverse experiences. Reported adverse experiences have not been serious. In the event of an overdose, symptomatic and supportive measures should be employed.

Rosuvastatin

There is no specific treatment in the event of overdose. In the event of overdose, the patient should be treated symptomatically and supportive measures instituted as required. Haemodialysis is unlikely to be of benefit.

9. CLINICAL STUDIES

In controlled clinical studies, Ezetimibe + Rosuvastatin (ROSUZET) (as coadministered ezetimibe and rosuvastatin) significantly reduced total cholesterol (total-C), low-density lipoprotein cholesterol (LDL-C), apolipoprotein B (Apo B), triglycerides (TG), and non-high-density lipoprotein cholesterol (non-HDL-C) and increased high-density lipoprotein cholesterol (HDL-C) in patients with hypercholesterolemia.

Primary Hypercholesterolemia**Ezetimibe + Rosuvastatin (ROSUZET)***Ezetimibe Add-on to On-going Rosuvastatin Therapy (Titration Studies)*

In a multicenter, randomized, double-blind, 6 week-actie comparator study (ACTE), 440 subjects at moderately high/high risk of coronary heart disease with LDL cholesterol levels failing to reach their LDL-C goal (100 mg/dL [<2.6 mmol/L] or 70 mg/dL [<1.8 mmol/L] depending on baseline characteristics) were stratified to treatment with rosuvastatin 5 mg or 10 mg for 4-5 weeks. Patients were then randomized to either doubling of their rosuvastatin dose (to 10 mg or 20 mg) or adding ezetimibe 10 mg to their rosuvastatin (5 or 10 mg) therapy, equivalent to Ezetimibe + Rosuvastatin (ROSUZET) 10/5 or 10/10.

Patients taking doses of ezetimibe and rosuvastatin equivalent to Ezetimibe + Rosuvastatin (ROSUZET) 10/5 or 10/10 achieved significantly greater LDL-C reductions compared to patients doubling the initial dose of rosuvastatin (to 10 mg or 20 mg) ($p < 0.001$). The LS mean percent change in LDL-C from baseline to the study end was -20.96% when ezetimibe 10 mg was added to rosuvastatin and -5.71% when the original rosuvastatin dose was doubled (data pooled across the rosuvastatin 5 mg and 10 mg strata). The LS mean treatment difference was -15.25% with a 95% CI (-19.89, -10.60) (Table 2).

Table 2: Analysis of Percent Change from Baseline in LDL-Cholesterol (mg/dL) at Study Endpoint After 6 Weeks of Treatment

Treatment	N	Baseline (mg/dL) Mean ± SD	Endpoint (mg/dL) Mean ± SD	LS mean (95% CI)	Difference in LS Mean (95% CI)	P value
Overall						
Ezetimibe + Rosuvastatin (ROSUZET) (10/5 or 10/10 mg)	21 9	103.9 ± 25.4	80.7 ± 32.3	-21.0 (-24.3, - 17.6)	-15.3 (-19.9, -10.6)	<0.001
Ezetimibe + Rosuvastatin (ROSUZET) (10 or 20 mg)	21 7	100.2 ± 24.4	92.9 ± 26.5	-5.71 (-9.05, - 2.38)		
Stratum I (Patients not at goal on rosuvastatin 5 mg)						
Ezetimibe + Rosuvastatin (ROSUZET) 10/5 mg	98	106.7 ± 23.5	85.9 ± 31.4	-18.0 (-22.69, - 13.15)	-12.3 (-18.9, -5.67)	<0.001
Rosuvastatin 10 mg	96	102.4 ± 23.4	95.4 ± 23.3	-5.61 (-10.4, - 0.79)		
Stratum II (Patients not at goal on rosuvastatin 10 mg)						
Ezetimibe + Rosuvastatin (ROSUZET) 10/10 mg	12 1	101.6 ± 26.7	76.8 ± 32.6	-23.7 (-28.3, - 19.1)	-17.5 (-23.9, -11.0)	<0.001
Rosuvastatin 20 mg	12 1	98.4 ± 25.2	90.9 ± 28.8	-6.28 (-10.9, - 1.69)		

Patients taking doses of ezetimibe and rosuvastatin equivalent to Ezetimibe + Rosuvastatin (ROSUZET) 10/5 or 10/10 significantly lowered total-cholesterol, non-HDL-C and Apo B, compared with doubling of the baseline dose of rosuvastatin ($p < 0.001$) and resulted in a significantly greater proportion of patients reaching LDL-C goal compared with doubling the baseline dose of rosuvastatin (10 mg or 20 mg) (59.4% vs. 30.9%; $p < 0.001$), where the LDL-C goal was <100 mg/dL (<2.6 mmol/L) for patients at moderately high risk/high risk for CHD without atherosclerotic vascular disease (AVD) and <70 mg/dL (<1.8 mmol/L) for patients at very high risk for CHD with AVD. In addition, there was a significantly greater proportion of patients reaching a LDL-C goal of <70 mg/dL (<1.8 mmol/L), regardless of risk status, in patients receiving Ezetimibe + Rosuvastatin (ROSUZET), compared with those receiving a doubling of

the baseline rosuvastatin dose (43.8% vs. 17.5%; $p < 0.001$). (See Figure 1.)

Figure 1a. Attainment of Pre-specified LDL-C Targets After 6 Weeks of Therapy

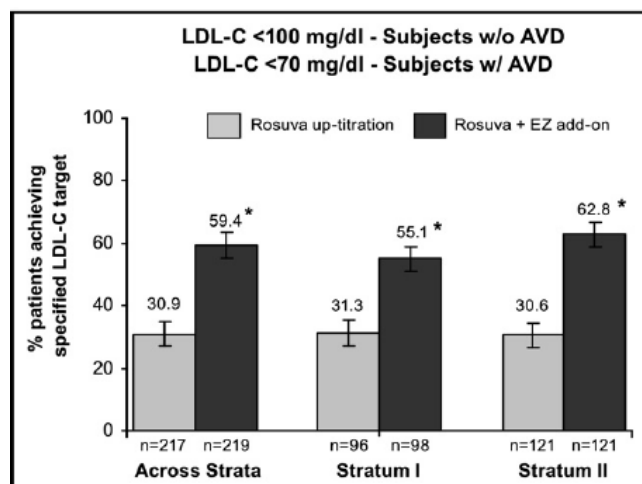
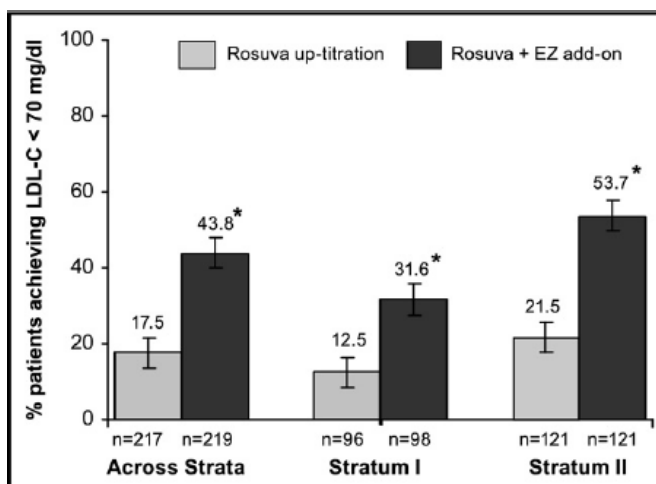


Figure 1b Attainment of LDL-C <70 mg/dL After 6 Weeks of Therapy



Across Strata = all ezetimibe 10 mg add-on to rosuvastatin (5 mg, 10 mg) versus all rosuvastatin up-titration (10 mg, 20 mg)

Stratum I = ezetimibe 10 mg add-on to rosuvastatin 5 mg versus rosuvastatin 10 mg

Stratum II = ezetimibe 10 mg add-on to rosuvastatin 10 mg versus rosuvastatin 20 mg.

* $p < 0.001$.

Study of Efficacy and Safety of the Ezetimibe/Rosuvastatin Fixed-dose Combination

In a multi-center, 8-week, double-blind, Phase 3 study, 412 Korean hypercholesterolemic subjects were randomized to treatment with a fixed-dose combination of ezetimibe 10 mg plus rosuvastatin at doses of 5 mg, 10 mg or 20 mg or rosuvastatin monotherapy at doses of 5 mg, 10 mg or 20 mg.

Patients taking the fixed-dose combination of ezetimibe plus rosuvastatin achieved significantly greater LDL-C reductions compared to patients treated with rosuvastatin alone pooled across doses ($p < 0.0001$) and at each dose comparison ($p \leq 0.01$) (Table 3).

Table 3: Percentage change from Baseline in LDL-Cholesterol (mg/dL) at Week 8 Full Analysis Set ANCOVA Result

LDL-C	Rosuvastatin	Ezetimibe/Rosuvastatin
Overall Effect		
LS Mean (SE)	-49.38 (1.86)	-59.10 (1.83)
LS Mean Difference (SE)		-9.73 (1.57)
95% CI for LS Mean		(-12.82, -6.63)

Difference		
p-value		<0.0001
Rosuvastatin 5 mg		
LS Mean (SE)	-42.45 (3.29)	-55.96 (3.26)
LS Mean Difference (SE)		-13.52 (2.77)
95% CI for LS Mean Difference		(-19.00, -8.04)
p-value		<0.0001
Rosuvastatin 10 mg		
LS Mean (SE)	-51.89 (3.13)	-58.57 (3.03)
LS Mean Difference (SE)		-6.67 (2.56)
95% CI for LS Mean Difference		(-11.75, -1.60)
p-value		0.0103
Rosuvastatin 20 mg		
LS Mean (SE)	-53.81 (2.95)	-62.82 (2.91)
LS Mean Difference (SE)		-9.01 (2.53)
95% CI for LS Mean Difference		(-14.01, -4.02)
p-value		0.0005

Pooled across doses, the fixed-dose combination of ezetimibe plus rosuvastatin lowered total cholesterol, non-HDL cholesterol, apolipoprotein B, and triglyceride levels more than rosuvastatin alone.

In addition, pooled across doses, treatment with ezetimibe/rosuvastatin resulted in a greater proportion of subjects reaching LDL-C goal compared with rosuvastatin alone (Table 4). The results for individual dose comparisons are shown in Table 5.

**Table 4: Proportion of Subjects Achieving LDL-C goals (Full Analysis Set)
Overall Population**

	Ezetimibe/Rosuvastatin (n=203)	Rosuvastatin (n=204)
Total patients achieving LDL-C goal, n (%)	191 (94.1) [†]	176 (86.3)

† p-value <.05 by the Cochran– Mantel– Haenszel test, with the CHD risk factors defined according to the National Cholesterol Education Program Adult Treatment Panel III.

Table 5: Proportion of Subjects Achieving LDL-C goals (Full Analysis Set) By Treatment Group

	Ezetimibe/ Rosuvastatin 10/5 mg	Rosuvastatin 5 mg	Ezetimibe/ Rosuvastatin 10/10 mg	Rosuvastatin 10 mg	Ezetimibe/ Rosuvastatin 10/20 mg	Rosuvastatin 20 mg
Total patients achieving LDL-C goal, n (%)	65 (97.0)†	50 (73.5)	62 (91.2)	63 (94.0)	64 (94.1)	63 (91.3)

† p-value <.05 by the Cochran– Mantel– Haenszel test, with the CHD risk factors defined according to the National Cholesterol Education Program Adult Treatment Panel III.

Ezetimibe

Monotherapy:

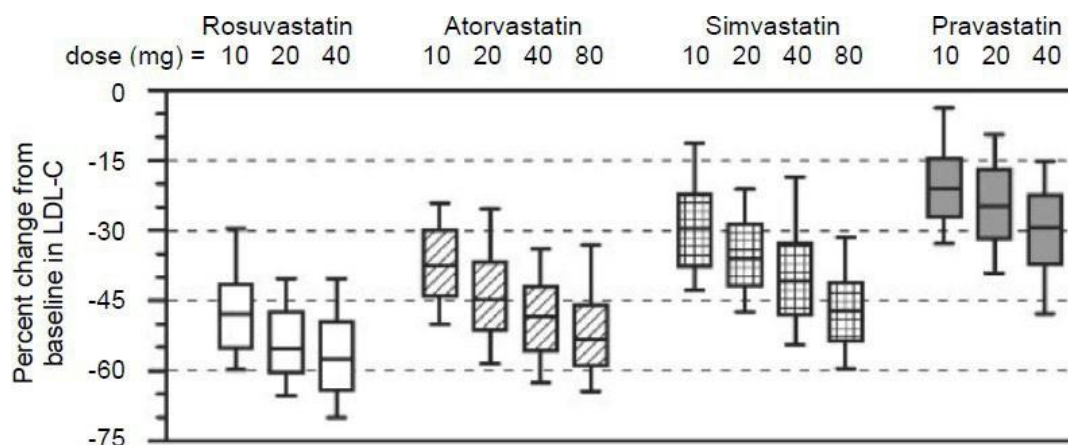
In two multicenter, double-blind, placebo-controlled, 12-week studies in 1719 patients with primary hypercholesterolemia, ezetimibe 10 mg significantly lowered total-C (-13%), LDL-C (-19%), Apo B (-14%), TG (-8%), and non-HDL-C (-17%) and increased HDL-C (+3%) compared to placebo. Reduction in LDL-C was consistent across age, sex, race, and baseline LDL-C.

Rosuvastatin

Active-Controlled Study:

Rosuvastatin was compared with the HMG-CoA reductase inhibitors atorvastatin, simvastatin, and pravastatin in a multicenter, open-label, dose-ranging study of 2240 patients with hyperlipidemia or mixed dyslipidemia. After randomization, patients were treated for 6 weeks with a single daily dose of either rosuvastatin, atorvastatin, simvastatin, or pravastatin (Figure 2).

Figure 2. Percent LDL-C Change by Dose of Rosuvastatin, Atorvastatin, Simvastatin, and Pravastatin at Week 6 in Patients with Hyperlipidemia or Mixed Dyslipidemia



Box plots are a representation of the 25th, 50th, and 75th percentile values, with whiskers representing the 10th and 90th percentile values. Mean baseline LDL-C: 189 mg/dL

JUPITER:

In the Justification for the Use of Statins in Primary Prevention: An Intervention Trial Evaluating Rosuvastatin (JUPITER) study, the effect of rosuvastatin on the occurrence of major atherosclerotic cardiovascular disease events was assessed in 17,802 men (≥ 50 years) and women (≥ 60 years). Patients were randomly assigned to placebo (n=8901) or rosuvastatin 20 mg once daily (n=8901) and were followed for a mean duration of 2 years.

The primary end point was a composite end point consisting of the time-to-first occurrence of any of the following major CV events: CV death, nonfatal myocardial infarction, nonfatal stroke, hospitalization for unstable angina or an arterial revascularization procedure.

Rosuvastatin significantly reduced the risk of major CV events (252 events in the placebo group vs. 142 events in the rosuvastatin group) with a statistically significant ($p < 0.001$) relative risk reduction of 44% and absolute risk reduction of 1.2% (see Figure 3). The risk reduction for the primary end point was consistent across the following predefined subgroups: age, sex, race, smoking status, family history of premature CHD, body mass index, LDL-C, HDL-C, and hsCRP levels.

The individual components of the primary end point are presented in Figure 4. Rosuvastatin significantly reduced the risk of nonfatal myocardial infarction, nonfatal stroke and arterial revascularization procedures. There were no significant treatment differences between the rosuvastatin and placebo groups for death due to

cardiovascular causes or hospitalizations for unstable angina.

Figure 3. Time to First Occurrence of Major Cardiovascular Events in JUPITER

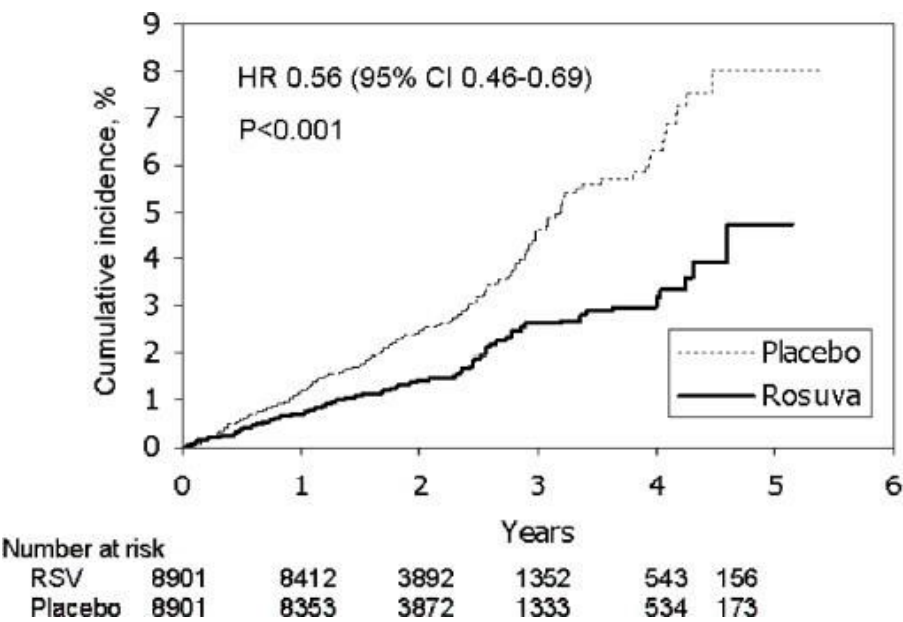
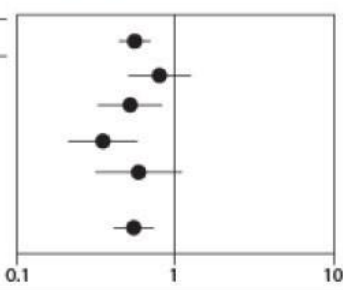


Figure 4. Major Cardiovascular Events by Treatment Group in JUPITER

End point	Number of events		HR (95% CI)	P value	Hazard Ratio (—95%—)
	Rosuva 20 mg (n=8901) n (rate*)	Placebo 20 mg (n=8901) n (rate*)			
Primary end point (MCE)	142 (7.6)	252 (13.6)	0.56 (0.46, 0.69)	<0.001	
Cardiovascular death**	35 (1.9)	44 (2.4)	0.80 (0.51, 1.24)	0.315	
Nonfatal Stroke	30 (1.6)	58 (3.1)	0.52 (0.33, 0.80)	0.003	
Nonfatal MI	22 (1.2)	62 (3.3)	0.35 (0.22, 0.58)	<0.001	
Hospitalized unstable Angina	16 (0.9)	27 (1.5)	0.59 (0.32, 1.10)	0.093	
Arterial revascularization	71 (3.8)	131 (7.1)	0.54 (0.41, 0.72)	<0.001	

* event rate/1000-patient years

** Cardiovascular death included fatal MI, fatal stroke, sudden death, and other adjudicated causes of CV death

The incremental benefit of Ezetimibe + Rosuvastatin (ROSUZET) on cardiovascular morbidity and mortality over and above that demonstrated for rosuvastatin has not been established.

Homozygous Familial Hypercholesterolemia (HoFH) Ezetimibe

A study was conducted to assess the efficacy of ezetimibe in the treatment of HoFH. This double-blind, randomized, 12-week study enrolled 50 patients with a clinical and/or genotypic diagnosis of HoFH, with or without concomitant LDL apheresis, already receiving atorvastatin or simvastatin (40 mg). Patients were randomized to one of three treatment groups, atorvastatin or simvastatin (80 mg), ezetimibe 10 mg administered with atorvastatin or simvastatin (40 mg), or ezetimibe 10 mg administered with atorvastatin or simvastatin (80 mg). Ezetimibe, administered with atorvastatin (40 or 80 mg) or simvastatin (40 or 80 mg), produced a reduction of LDL-C of 21% from baseline compared with increasing the dose of simvastatin or atorvastatin monotherapy from 40 to 80 mg, which resulted in a reduction of LDL-C of 7% from baseline. In those treated with ezetimibe plus 80mg atorvastatin or with ezetimibe plus 80mg simvastatin, LDL-C was reduced by 27%.

Rosuvastatin

In an open-label, forced-titration study, HoFH patients (n=40, 8-63 years) were evaluated for their response to rosuvastatin 20 to 40 mg titrated at a 6-week interval. In the overall population, the mean LDL-C reduction from baseline was 22%. About one-third of the patients benefited from increasing their dose from 20 mg to 40 mg with further LDL lowering of greater than 6%. In the 27 patients with at least a 15% reduction in LDL-C, the mean LDL-C reduction was 30% (median 28% reduction). Among 13 patients with an LDL-C reduction of <15%, 3 had no change or an increase in LDL-C. Reductions in LDL-C of 15% or greater were observed in 3 of 5 patients with known receptor negative status.

10. CLINICAL PHARMACOLOGY

10.1 Therapeutic Class

Ezetimibe + Rosuvastatin (ROSUZET) is a lipid-lowering product that selectively inhibits the intestinal absorption of cholesterol and related plant sterols and inhibits the endogenous synthesis of cholesterol.

10.2 Mechanism of Action

Ezetimibe + Rosuvastatin (ROSUZET)

Plasma cholesterol is derived from intestinal absorption and endogenous synthesis. Ezetimibe + Rosuvastatin (ROSUZET) contains ezetimibe and rosuvastatin, two lipid-lowering compounds with complementary mechanisms of action. Ezetimibe + Rosuvastatin (ROSUZET) reduces elevated total-C, LDL-C, Apo B, TG, and non-HDL-C, and increases HDL-C through dual inhibition of cholesterol absorption and synthesis.

Ezetimibe

Ezetimibe inhibits the intestinal absorption of cholesterol. Ezetimibe is orally active and has a mechanism of action that differs from other classes of cholesterol-reducing compounds (e.g., statins, bile acid sequestrants [resins], fibric acid derivatives, and plant stanols). The molecular target of ezetimibe is the sterol transporter, Niemann-Pick C1-Like 1 (NPC1L1), which is responsible for the intestinal uptake of cholesterol and phytosterols.

Ezetimibe localizes at the brush border of the small intestine and inhibits the absorption of cholesterol, leading to a decrease in the delivery of intestinal cholesterol to the liver; statins reduce cholesterol synthesis in the liver and together these distinct mechanisms

provide complementary cholesterol reduction.

In a 2-week clinical study in 18 hypercholesterolemic patients, ezetimibe inhibited intestinal cholesterol absorption by 54%, compared with placebo.

A series of preclinical studies was performed to determine the selectivity of ezetimibe for inhibiting cholesterol absorption. Ezetimibe inhibited the absorption of [^{14}C]-cholesterol with no effect on the absorption of triglycerides, fatty acids, bile acids, progesterone, ethinyl estradiol, or the fat-soluble vitamins A and D.

Rosuvastatin

Rosuvastatin is a selective, potent and competitive inhibitor of HMG-CoA reductase, the rate-limiting enzyme that converts 3-hydroxy-3-methylglutaryl coenzyme A to mevalonate, a precursor of cholesterol. Triglycerides (TG) and cholesterol in the liver are incorporated, with apolipoprotein B (ApoB), into very low density lipoprotein (VLDL) and released into the plasma for delivery to peripheral tissues. VLDL particles are TG-rich. Cholesterol-rich low density lipoprotein (LDL) is formed from VLDL and is cleared primarily through the high affinity LDL receptor in the liver.

Rosuvastatin produces its lipid-modifying effects in two ways; it increases the number of hepatic LDL receptors on the cell-surface, enhancing uptake and catabolism of LDL and it inhibits the hepatic synthesis of VLDL, thereby reducing the total number of VLDL and LDL particles. High density lipoprotein (HDL), which contains ApoA-I is involved, amongst other things, in transport of cholesterol from tissues back to the liver (reverse cholesterol transport).

The involvement of LDC-C in atherogenesis has been well documented. Epidemiological studies have established that high LDL-C, TG, low HDL-C and ApoA-I have been linked to a higher risk of cardiovascular disease. Intervention studies have shown the benefits in mortality and CV event rates of lowering LDL-C and TG or raising HDL-C. More recent data has linked the beneficial effects of HMG CoA reductase inhibitors to lowering of non-HDL (i.e. all circulating cholesterol not in HDL) and ApoB or reducing the ApoB/ApoA-I ratio.

10.3 Pharmacokinetics General Introduction

Ezetimibe + Rosuvastatin (ROSUZET)

Ezetimibe + Rosuvastatin (ROSUZET) has been shown to be bioequivalent to coadministration of corresponding doses of ezetimibe and rosuvastatin tablets.

Absorption

Ezetimibe

After oral administration, ezetimibe is rapidly absorbed and extensively conjugated to a pharmacologically active phenolic glucuronide (ezetimibe-glucuronide). Mean maximum plasma concentrations (C_{max}) occur within 1 to 2 hours for ezetimibe-glucuronide and 4 to 12 hours for ezetimibe. The absolute bioavailability of ezetimibe cannot be determined as the compound is virtually insoluble in aqueous media suitable for injection.

Concomitant food administration (high fat or non-fat meals) had no effect on the oral

bioavailability of ezetimibe when administered as ezetimibe 10-mg tablets.

Rosuvastatin

Rosuvastatin is administered orally in the active form with peak plasma levels occurring 5 hours after dosing. Exposure increases linearly over the dose range. Absolute bioavailability is 20%. There is minimal accumulation on repeated once daily dosing.

Distribution

Ezetimibe

Ezetimibe and ezetimibe-glucuronide are bound 99.7% and 88 to 92% to human plasma proteins, respectively.

Rosuvastatin

Rosuvastatin is approximately 90% bound to plasma proteins, mostly albumin. The parent compound, accounts for greater than 90% of the circulating active HMG CoA reductase inhibitor activity.

Metabolism

Ezetimibe

Ezetimibe is metabolized primarily in the small intestine and liver via glucuronide conjugation (a phase II reaction) with subsequent biliary excretion. Minimal oxidative metabolism (a phase I reaction) has been observed in all species evaluated. Ezetimibe and ezetimibe-glucuronide are the major drug-derived compounds detected in plasma, constituting approximately 10 to 20 % and 80 to 90 % of the total drug in plasma, respectively. Both ezetimibe and ezetimibe- glucuronide are slowly eliminated from plasma with evidence of significant enterohepatic recycling. The half-life for ezetimibe and ezetimibe-glucuronide is approximately 22 hours.

Rosuvastatin

Rosuvastatin undergoes first pass extraction in the liver, which is the primary site of cholesterol synthesis and LDL-C clearance.

Elimination

Ezetimibe

Following oral administration of ¹⁴C-ezetimibe (20 mg) to human subjects, total ezetimibe accounted for approximately 93% of the total radioactivity in plasma. Approximately 78% and 11% of the administered radioactivity were recovered in the feces and urine, respectively, over a 10-day collection period. After 48 hours, there were no detectable levels of radioactivity in the plasma.

Rosuvastatin

Rosuvastatin undergoes limited metabolism (approximately 10%), mainly to the N-desmethyl form, and 90% is eliminated as unchanged drug in the feces with the remainder being excreted in the urine.

Special Populations

Renal Impairment

Ezetimibe

After a single 10-mg dose of ezetimibe in patients with severe renal disease (n=8; mean CrCl $\leq$ 30 mL/min/1.73 m²), the mean AUC for total ezetimibe was increased approximately 1.5-fold, compared to healthy subjects (n=9).

An additional patient in this study (post-renal transplant and receiving multiple medications, including cyclosporine) had a 12-fold greater exposure to total ezetimibe.

Rosuvastatin

In a study in subjects with varying degrees of renal impairment, mild to moderate renal disease had little influence on plasma concentrations of rosuvastatin. However, subjects with severe impairment (CrCl <30 mL/min) had a 3-fold increase in plasma concentration compared to healthy volunteers.

Hepatic Impairment***Ezetimibe***

After a single 10-mg dose of ezetimibe, the mean area under the curve (AUC) for total ezetimibe was increased approximately 1.7-fold in patients with mild hepatic impairment (Child-Pugh score 5 or 6), compared to healthy subjects. In a 14-day, multiple-dose study (10 mg daily) in patients with moderate hepatic impairment (Child-Pugh score 7 to 9), the mean AUC for total ezetimibe was increased approximately 4-fold on Day 1 and Day 14 compared to healthy subjects. No dosage adjustment is necessary for patients with mild hepatic impairment. Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe (Child-Pugh score >9) hepatic impairment, ezetimibe is not recommended in these patients [see 4 WARNINGS AND PRECAUTIONS, 4.4 Hepatic Impairment].

Rosuvastatin

In a study in subjects with varying degrees of hepatic impairment there was no evidence of increased exposure to rosuvastatin other than in the 2 subjects with the most severe liver disease (Child-Pugh scores of 8 and 9). In these subjects systemic exposure was increased by at least 2-fold compared to subjects with lower Child-Pugh scores.

Pediatric***Ezetimibe***

The pharmacokinetics of ezetimibe are similar between children $\geq$ 6 years and adults. Pharmacokinetic data in the pediatric population $<$ 6 years of age are not available.

Rosuvastatin

The exposure in children and adolescents with heterozygous familial hypercholesterolemia appears to be similar to or lower than that in adult patients with dyslipidemia.

Geriatric***Ezetimibe***

Plasma concentrations for total ezetimibe are about 2-fold higher in the elderly ($\geq$ 65 years) than in the young (18 to 45 years). LDL-C reduction and safety profile are comparable between elderly and young subjects treated with ezetimibe.

Rosuvastatin

There was no clinically relevant effect of age on the pharmacokinetics of rosuvastatin in adults.

Race

Ezetimibe

Based on a meta analysis of pharmacokinetic studies with ezetimibe, there were no pharmacokinetic differences between Blacks and Caucasians.

Rosuvastatin

Pharmacokinetic studies show an approximate 2-fold elevation in median AUC in Asian subjects compared with Caucasians.

A population pharmacokinetic analysis revealed no clinically relevant differences in pharmacokinetics among Caucasian, Hispanic and Black or Afro-Caribbean groups.

Sex

Ezetimibe

Plasma concentrations for total ezetimibe are slightly higher (<20 %) in women than in men. LDL-C reduction and safety profile are comparable between men and women treated with ezetimibe.

Rosuvastatin

There was no clinically relevant effect of sex on the pharmacokinetics of rosuvastatin in adults.

Genetic polymorphisms

Disposition of HMG-CoA reductase inhibitors, including rosuvastatin, involves OATP181 and BCRP transporter proteins. In patients with SLCO181 (OATP181) and/or ABCG2 (BCRP) genetic polymorphisms there is a risk of increased rosuvastatin exposure. Individual polymorphisms of SLCO181 c.521 CC and ABCG2 c.421AA are associated with an approximate 1.6-fold higher rosuvastatin exposure (AUC) or 2.4-fold higher exposure, respectively, compared to the SLCO1 B1 c.521 TT or ABCG2. c.421 CC genotypes.

11. PRECLINICAL SAFETY DATA

Rosuvastatin

Preclinical data reveal no special hazards for humans based on conventional studies of safety pharmacology, repeat-dose toxicity, genotoxicity, carcinogenic potential and reproductive toxicity.

12. DOSAGE FORMS AND PACKAGING AVAILABLE

Each EZETIMIBE + ROSUVASTATIN (ROSUZET) 10 mg/10 mg film-coated tablet contains 10 mg of ezetimibe and 10.4 mg of rosuvastatin calcium, equivalent to 10 mg of rosuvastatin, available in blister pack of 10's, box of 30's.

Each EZETIMIBE + ROSUVASTATIN (ROSUZET) 10 mg/20 mg film-coated tablet contains 10 mg of ezetimibe and 20.8 mg of rosuvastatin calcium, equivalent to 10 mg

of rosuvastatin, available in blister pack of 10's, box of 30's.

13. PHARMACEUTICAL FORM

10 mg/10mg: Pale yellowish pink oval tablet debossed with "R10" on one face

10 mg/20mg: Pink oval tablet debossed with "R20" on one face

14. PHARMACEUTICAL PARTICULARS

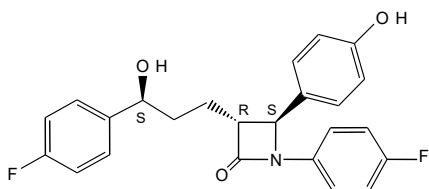
14.1 Chemistry

Ezetimibe + Rosuvastatin (ROSUZET) contains ezetimibe, a selective inhibitor of intestinal cholesterol and related phytosterol absorption, and rosuvastatin, a 3-hydroxy-3-methylglutaryl- coenzyme A (HMG-CoA) reductase inhibitor.

Ezetimibe

The chemical name of ezetimibe is 1-(4-fluorophenyl)-3(R)-[3-(4-fluorophenyl)-3(S)-hydroxypropyl]-4(S)-(4-hydroxyphenyl)-2-azetidinone. The empirical formula is C₂₄H₂₁F₂NO₃. Its molecular weight is 409.4.

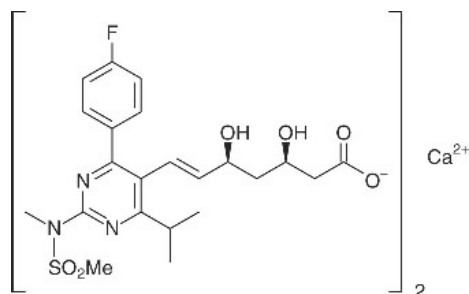
Ezetimibe is a white, crystalline powder that is freely to very soluble in ethanol, methanol, and acetone and practically insoluble in water. Its structural formula is:



Rosuvastatin

Rosuvastatin calcium is a synthetic lipid-lowering agent for oral administration. The chemical name for rosuvastatin calcium is calcium bis[(3R,5S,6E)-7-[4-(4-fluorophenyl)-6-(1-methylethyl)- 2-[methyl(methylsulfonyl)amino]pyrimidin-5-yl]-3,5-dihydroxyhept-6-enoate]. The empirical formula for rosuvastatin calcium is (C₂₂H₂₇FN₃O₆S)₂Ca. Its molecular weight is 1001.14.

Rosuvastatin calcium is a white or almost white powder that is soluble in methylene chloride. It is slightly soluble in water. Rosuvastatin calcium is a slightly hygroscopic compound with a partition coefficient (octanol/water) of 4.55 at pH of 3.82. Its structural formula is:



14.2 Composition

Active Ingredients

Ezetimibe + Rosuvastatin (ROSUZET) is available for oral use as tablets containing 10 mg of ezetimibe and: 10.4 mg of rosuvastatin calcium, equivalent to 10 mg of rosuvastatin (Ezetimibe + Rosuvastatin (ROSUZET 10/10)); 20.8 mg of rosuvastatin calcium, equivalent to 20 mg of rosuvastatin (Ezetimibe + Rosuvastatin (ROSUZET 10/20)).

Inactive Ingredients (List of excipients)

Each tablet of Ezetimibe + Rosuvastatin (ROSUZET) contains the following inactive ingredients: lactose monohydrate, sodium lauril sulfate, croscarmellose sodium, povidone, cellulose microcrystalline, crospovidone, magnesium stearate, and mannitol.

The film coating contains: hypromellose, polyethylene glycol 6000/macrogol 8000, talc, titanium oxide, iron oxide yellow and iron oxide red.

14.3 Storage

Store at temperatures not exceeding 30°C. Keep product in the original blister package until use, to protect from light. Keep Ezetimibe + Rosuvastatin (ROSUZET) and all medicines safely away from children.

15. EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

Pharmacology testing revealed no evidence of a sedative effect of rosuvastatin. From the safety profile, rosuvastatin is not expected to adversely affect the ability to drive or use machines.

16. MANUFACTURED BY

Hanmi Pharmaceutical Co., Ltd.
214, Muha-ro, Paltan-myeon, Hwaseong-si Gyeonggi-do, Republic of Korea

17. PRODUCT REGISTRATION HOLDER

Organon Malaysia Sdn. Bhd.
3A-08-02, Level 8, Corporate Tower 3A, Pavilion Damansara Heights,

No. 3, Jalan Damanlela 50490 Kuala Lumpur, Malaysia.

18. DATE OF REVISION

12 November 2025

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